

L19.pdf
PDF

Generate a complete lecture scribe intended to be used as authoritative reference material for exam preparation in a closed-notes setting.

The scribe must be constructed exclusively from the provided lecture slides and relevant textbook material. All content in the output must be directly traceable to this context, and no additional explanations, assumptions, examples, or interpretations may be introduced.

The objective of the scribe is to accurately reproduce the lecture content in a form that enables precise revision of all concepts, results, and problem-solving procedures.

The lecture scribe should:

- Preserve the original structure and ordering of topics as presented in the lecture.
- Include all definitions, notations, and formal terminology exactly as introduced.
- Clearly state all assumptions, conditions, and constraints where applicable.
- Present all theorems, propositions, and formal results in their complete form.
- Reconstruct all proofs and derivations step by step, including all intermediate steps shown in the lecture.
- Include all worked examples and solved problems with full procedural detail as given in the lecture material.
- Maintain strict mathematical correctness and consistency throughout.
- Use well-defined sections and headings to reflect the logical flow of the lecture content.

The lecture scribe must not:

- Add intuitive explanations, commentary, or simplifications beyond what is explicitly stated in the provided material.
- Introduce new examples, alternative methods, or additional results.
- Omit intermediate reasoning steps that are present in the lecture.
- Alter the logical sequence or organization of the lecture content.

The final output must function as a faithful reconstruction of the lecture, suitable for direct use in exam revision without reliance on external sources.

Below is a **faithful lecture scribe** reconstructed strictly from the provided slides of *Lecture 19: N-Variate Joint PDF/PMF/CDF*. All content is preserved in structure, notation, and sequence, without additional interpretation.

CSE400 – Fundamentals of Probability in Computing

Lecture 19: N-Variate Joint PDF/PMF/CDF L19

1. Joint Distribution of N Random Variables

1.1 Representation of Multiple Random Variables

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For random variables $X_1, X_2, \dots, X_N$, the vector form is:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- The joint distribution describes the probability behavior of all variables together.
- It can be represented using:
 - Joint PMF
 - Joint PDF
 - Joint CDF  L19

2. Jointly Gaussian Random Variables

2.1 Joint Gaussian PDF (Two Variables)

For two variables X and Y :

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

Where:

- μ_X, μ_Y : means
- σ_X, σ_Y : standard deviations
- ρ : Pearson correlation coefficient  L19

3. Example 1: Jointly Gaussian Variables

Given

$$\mu_X = 2, \quad \sigma_X^2 = 1$$

$$\mu_Y = -3, \quad \sigma_Y^2 = 4$$

$$\text{cov}(X, Y) = -1$$

(a) Joint PDF Construction**Step 1: Correlation Coefficient**

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

$$\rho_{XY} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

Step 2: Substitute into General Formula

$$\mu_X = 2, \sigma_X = 1, \mu_Y = -3, \sigma_Y = 2, \rho = -0.5$$

Final Joint PDF

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{2}{3} \left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4}\right]\right)$$

(b) Marginal PDFs

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

Key Property

If X and Y are jointly Gaussian:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

Marginal Results

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-2)^2}{2}\right)$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(y+3)^2}{8}\right)$$

(c) Probability $\Pr(X < 0)$ **Step 1: Identify CDF**

$$F_X(x) = \Pr(X \leq x), \quad \Pr(X < 0) = F_X(0)$$

Step 2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$

$$\Pr(X < 0) = \Pr\left(Z < \frac{0 - 2}{1}\right) = \Pr(Z < -2)$$

Step 3: Symmetry Property

$$\phi(-x) = 1 - \phi(x) = \Pr(Z > x)$$

$$\Pr(Z < -2) = \Pr(Z > 2)$$

Step 4: Q-Function

$$Q(x) = \Pr(Z > x)$$

Final Answer

$$\Pr(X < 0) = Q(2)$$

4. Application: Large MIMO System

System Setup

- M : transmit antennas
- N : receive antennas

Channel Coefficients

Each path between antenna i and j :

$$h_{ij}$$

is modeled as a random variable.

Channel Matrix

$$H = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1N} \\ h_{21} & h_{22} & \cdots & h_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ h_{M1} & h_{M2} & \cdots & h_{MN} \end{bmatrix}_{M \times N}$$

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5. Joint PMF

For discrete random variables:

$$P_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 = x_1, \dots, X_N = x_N)$$

□ L19

6. Joint CDF

$$F_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 \leq x_1, \dots, X_N \leq x_N)$$

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7. Joint PDF

$$f_{X_1, \dots, X_N}(x_1, \dots, x_N) = \frac{\partial^N}{\partial x_1 \dots \partial x_N} F_{X_1, \dots, X_N}(x_1, \dots, x_N)$$

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8. Independence of Random Variables

Condition

$$\mathbf{X} = [X_1, X_2, \dots, X_N]^T$$

If independent:

$$f_X(x_1, \dots, x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

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9. Example 2: Uniformly Distributed 3D Vector

Given

$$X = (X_1, X_2, X_3)$$

with:

$$|X_1| \leq 1, |X_2| \leq 1, |X_3| \leq 1$$

$$f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \begin{cases} C, & \text{inside cube} \\ 0, & \text{otherwise} \end{cases}$$

(a) Find C

$$\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 C \, dx_1 dx_2 dx_3 = 1$$

$$C(2)(2)(2) = 1 \Rightarrow C = \frac{1}{8}$$

(b) Marginal $f_{X_1, X_2}(x_1, x_2)$

$$f_{X_1, X_2}(x_1, x_2) = \int_{-1}^1 \frac{1}{8} dx_3 = \frac{1}{4}$$

for $-1 \leq x_1, x_2 \leq 1$

(c) Marginal $f_{X_1}(x_1)$

$$f_{X_1}(x_1) = \int_{-1}^1 \int_{-1}^1 \frac{1}{8} dx_2 dx_3 = \frac{1}{2}$$

for $-1 \leq x_1 \leq 1$

(d) Conditional PDF

$$f_{X_1|X_2, X_3}(x_1|x_2, x_3) = \frac{f_{X_1, X_2, X_3}(x_1, x_2, x_3)}{f_{X_2, X_3}(x_2, x_3)} = \frac{1/8}{1/4} = \frac{1}{2}$$

(e) Conditional PDF

$$f_{X_1, X_2|X_3}(x_1, x_2|x_3) = \frac{f_{X_1, X_2, X_3}}{f_{X_3}} = \frac{1/8}{1/2} = \frac{1}{4}$$

(f) Independence Check

$$\begin{aligned} \text{LHS} &= f_{X_1, X_2, X_3} = \frac{1}{8} \\ \text{RHS} &= f_{X_1} f_{X_2} f_{X_3} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \end{aligned}$$

Conclusion

$$\text{LHS} = \text{RHS} \Rightarrow \text{independent}$$

10. Example 3: Trivariate Random Variables

Given

$$f_{XYZ}(x, y, z) = \begin{cases} K e^{-(ax+by+cz)}, & x, y, z > 0 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find K

$$\begin{aligned} \int_0^\infty \int_0^\infty \int_0^\infty K e^{-(ax+by+cz)} dx dy dz &= 1 \\ K \left(\frac{1}{a}\right) \left(\frac{1}{b}\right) \left(\frac{1}{c}\right) &= 1 \\ K &= abc \end{aligned}$$

(b) Marginal Joint PDF $f_{XY}(x, y)$

$$f_{XYZ}(x, y) = \int_0^\infty abc e^{-(ax+by+cz)} dz$$

$$= abe^{-(ax+by)}, \quad x, y > 0$$

(c) Marginal PDF $f_X(x)$

$$f_X(x) = \int_0^\infty abe^{-(ax+by)} dy = ae^{-ax}, \quad x > 0$$

(d) Independence Check

Condition

$$f_{XYZ}(x, y, z) = f_X(x)f_Y(y)f_Z(z)$$

LHS

$$abce^{-(ax+by+cz)}$$

RHS

$$(ae^{-ax})(be^{-by})(ce^{-cz}) = abce^{-(ax+by+cz)}$$

Conclusion

$$\text{LHS} = \text{RHS} \Rightarrow X, Y, Z \text{ are independent}$$

End of Lecture 19



Sources

